

CS2610: Computer Organization and Architecture Lab

Lab 4: Logisim Basics

Instructions

The purpose of this assignment is to become familiar with the Logisim interface. The library reference in the Help menu contains detailed documentation for modifying circuit elements and wiring, and for features like defining subcircuits.

You have been provided 4 tasks. You will be required to design circuits for the tasks and report simulation results. Submit a compressed folder with

- Logisim files for all completed circuits
- a .PDF file containing your answers and all necessary verification screenshots

The filenames of the folder and the PDF should be your roll number. The names of the circuit files should be 'Task1.circ', 'Task2.circ', and so on.

Maximum marks: 20. Deadline: **Monday March 2, 5 PM.**

Task 1 (5 marks)

You have to design a circuit which implements the function

$$F(A, B, C) = A\bar{B} + BC$$

- Write the truth tables for the function.
- Design the circuit.
- Verify $F(1, 1, 0)$.

Task 2 (5 marks)

Make a 4-bit ripple-carry adder.

- Design the circuit.
- Verify $1001 + 0011$.

Task 3 (5 marks)

Design a 4 bit register implemented using D flip-flops. The register holds its value unless the enable (EN) signal is active, in which case it loads from the input.

- Design the circuit.
- Load 1000 and verify holding behaviour for additional two cycles.

Task 4 (5 marks)

Design a 2-bit saturating counter. Start with a 2-bit number implemented with D flip flops and a clock source.

- Derive the next-state equations for each bit.
- Design the circuit and verify your logic by simulating 5 cycles.